

Lab Sheet 5

1. Define name=['AA','BB','CC','DD','EE','FF','ZZ','YY','XX','AD','KK','MM']
 - a. Print length of the list
 - b. Change the name DD to GG and print the new list without correcting the list.
 - c. Print the following list items: BB, EE, YY, KK using list slicing.
2. Define num=[12, 23, 2, 4, -6, 67, 45, 30, 21, 3, 5, 4, 9, 14, 10, 22, 89].
 - a. Print index of 5
 - b. Print the index of 4 that lies between 5 and 9.
 - c. Add the value 101 at the end.
 - d. Insert 27 in the 7th index.
 - e. Sort the numbers in descending order.
 - f. Sort the numbers in ascending order.
 - g. Remove element 21 and print the new list.
 - h. Remove an element from the 12th index.
 - i. Reverse your list
3. Write a Python program to sum all the items in the list [23, 34, -3, 55, 34, 12, 2, 6]
4. Write a Python program to multiply all the items in a list [2, 6, 12, 13, -4, 2.5, 8]
5. Write a Python program to get the largest number from a list [1, 7, 10, 34, 2, 8].
6. Create the following lists: [0,0,0,0,0,0], [1,3,5,7,9] and [1,2,4,8,16,32,64,128,256] without listing elements explicitly. (*Hint – list comprehension – use a for loop in the list.*)
7. Write a python program to replace the last element in a list with another list.
Sample data: [1, 3, 5, 7, 9, 10], [2, 4, 6, 8]. Expected Output: [1, 3, 5, 7, 9, 2, 4, 6, 8]
8. Write a Python program to move all zero digits to the end of a given list of numbers.
Original list: [3, 4, 0, 0, 0, 6, 2, 0, 6, 7, 6, 0, 0, 0, 9, 10, 7, 4, 4, 5, 3, 0, 0, 2, 9, 7, 1]
Expected output: [3, 4, 6, 2, 6, 7, 6, 9, 10, 7, 4, 4, 5, 3, 2, 9, 7, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]
9. Write a Python program to make a list of 50 random integers between 0 and 20
 - a. Print out the number of times each number is repeated in the list. This may be done by creating a new list with the index as the number and the count of each number as elements.
 - b. Find the average of all the numbers in the list.
 - c. Remove all the duplicate values in the list
10. Write a program to accept a word as input and make a list of all the letters in the word. Avoid duplication of letters.